



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT / DATABASE DESIGN AND DEVELOPMENT / DATABASE DESIGN AND APPLICATION (PAPER 1)
COURSE CODE	:	ICT450/450/ITS432
EXAMINATION	:	FEBRUARY 2022
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of **FIVE (5)** questions.
2. Answer **ALL** questions in English.
3. Make sure your answer file is rename as **ICT450_FinalTest_StudentID**.
Example : ICT450_NorIntanShafini_2022123456
4. Please submit **ONLY ONE (1)** answer file using PDF format.
5. This is an open book assessment, however candidates are **NOT ALLOWED** to discuss with anybody. **0 mark** will be given if cheating is detected in your answer file.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

QUESTION 1

a) Database is the best way to store and manage data. Identify **FOUR (4)** data that will be involved in online purchasing.

(4 marks)

b) Describe the basic features of the relational data model and discuss their importance to the end user and the designer.

(4 marks)

c) Table 1, 2 and 3 show the data in a local IT learning academy that is famous for various computer learning courses. Because of its popularity, there are increasing number of students enrol in the academy. Since then, the academy needs an efficient automated data storage, hence you are required to determine relational keys to ensure each row in the table is uniquely identifiable.

Table 1: **STUDENT** table

STUDENT_IC	STUDENT_ID	FIRST_NAME	LAST_NAME	EMAIL	COURSE_ID
80193939979	202587788	Ahmad	Ali	AA@gmail.com	40
40122566748	202575526	Haqiem	Rosly	haqiem@gmail.com	40
20164723670	202552244	Wany	Hazwan	wany@gmail.com	50
10108990074	202536987	Haliza	Husni	Ha@gmail.com	50
80116756736	202512369	Heechul	Kim	heechul@gmail.com	40
90137476323	202533969	James	Reece	james@gmail.com	20
50143670971	202574563	Canos	Sergi	Ola21@gmail.com	20

Table 2: **COURSE** table

COURSE_ID	COURSE_NAME
10	Networking
20	Software Developer
30	Web Designer
40	Database Developer
50	Data Analyst
60	IT Professional

Table 3: **STAFF** table

STAFF_ID	STAFF_NAME	COURSE_ID
101	Ronaldo	10
102	Vardy	30
103	Toney	60
104	Jimenez	20
105	Kane	40
106	Lukaku	50

i) Identify **THREE (3)** possible superkeys, **TWO (2)** candidate keys and primary key(s) for the STUDENT table. Justify your answer.

(6 marks)

ii) Do the tables display both entity integrity and referential integrity? Answer YES or NO, and then explain your answer for each table.

(6 marks)

QUESTION 2

a) Database Design phase is the second phase of the Database Life Cycle (DBLC) that focuses on the design of the database model. It consists of three essential stages which are conceptual design, logical design and physical design. Briefly explain the differences among these stages and identify either they are software-dependent or not.

(8 marks)

b) Describe the **FOUR (4)** characteristics of big data. Give an example for each characteristic.

(8 marks)

c) Spreadsheet application such as Microsoft Excel (shown in Figure 1) can also be used to store data. However, spreadsheet application cannot be considered as a database system because there are several basic database functions that a spreadsheet cannot perform. Explain **TWO (2)** features of spreadsheet application that are different from database system.

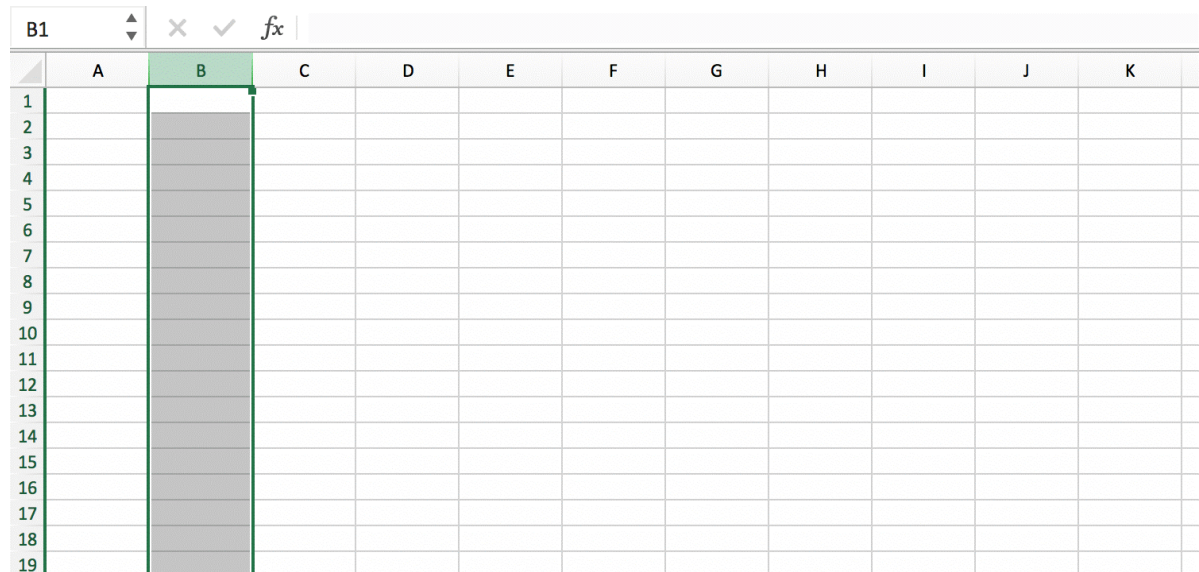


Figure 1: Spreadsheet Application

(4 marks)

QUESTION 3

Table 4 shows the reservation data for a dinner at a Queen Hotel.

Table 4: Reservation data

CustomerID	A01	A01	A02	A03
Reservation ID	R1	R3	R2	R4
Reservation Date	19/02/2022	20/02/2022	19/02/2022	23/02/2022
Time Slot	8:00 PM	8:00 PM	9:00 PM	7:00 PM
First Name	Intan		Fatin	Masitah
Last Name	Nasa		Ruslan	Mat
Email	intan@gmail.com	intan@gmail.com	fatin@gmail.com	mas@gmail.com
Phone Number	017-12345678			012-3455777
Number of Guests	2	3	1	1
Staff In Charge	111 – Yusri	222 – Fashiha	111 – Yusri	333 - Laila
Type Of Cuisine	Italian	Japanese	Italian	Chinese
Menu ID and Name	IT1 – Lasagna IT2 - Pizza	J1 - Gyoza J2 - Sashimi	IT3 - Tiramisu	C1 - Kung Pao Chicken C2 - Fried Noodles

Answer the following questions.

- a) State the normal form of the above table. Explain your answer with reasons.
(3 marks)
- b) Convert the table into a relational schema. Identify the primary key(s) and partial dependencies and/or transitive dependencies.
(6 marks)
- c) Normalize the relational schema in (b) until third normal form (3NF). Show all your work.
(11 marks)

QUESTION 4

Consider the following relational database:

PUBLISHER (PublisherName, City, State, Country, PhoneNum)

AUTHOR (AuthorNum, AuthorName)

BOOK (ISBN, BookName, PublicationYear, AuthorNum, PublisherName)

CUSTOMER (CustNum, CustName, Street, City, State, Country, Email)

ORDER (OrderID, ISBN, CustNum, Date, Quantity)

- a) Create table BOOK.
(4 marks)
- b) Display the book number, book name and author name of all the books published by John Wiley & Sons Inc and Pustaka Berhad. View the results by book name in descending order.
(4 marks)
- c) Find how many books that have been published in 2021 by Pustaka Trading located in Jasin, Melaka.
(4 marks)
- d) Find the publisher which successfully published more than 50 books
(4 marks)

e) Find the book name which is total order is more than 3.

(4 marks)

QUESTION 5

UiTM is planning to develop a volunteer management system for a better coordination and reporting for the volunteerism activities. As a database designer, you are assigned to design the database system according to the following requirements:

- Any student or staff can be a volunteer.
- Each student or staff can fill only one entry form. The details are volunteer ID, register date and total hours of working.
- Volunteer can volunteer in more than one projects.
- Each assigned project will have details about start date, end date, hours, station and status. Each assigned project also will lead by one faculty.
- Each project will have their own project ID and project name such as trash clean up, money donation, animal shelter and others.

Based on the given information:

a) Construct an Entity Relationship Diagram (ERD) complete with all of the required tables, attributes and relationships using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

b) Identify **TWO (2)** relevant information that you can get from the ERD you construct in (a). Briefly explain your answer.

(4 marks)

END OF QUESTION PAPER